

GENERAL SYSTEMS THEORY

~~Systems Analysis & Design~~

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FRANCISCO JOSÉ DE CALDAS

Outline

- 1 Basic Concepts ✓
- 2 Information Theory and Entropy ✓
- 3 Graphs and Networks Theory ✓
- 4 Paradigms Supporting GST ✓



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What is General Systems Theory?

- General Systems Theory (**GST**) is an interdisciplinary framework for understanding and analyzing complex systems.
- It was introduced by **Ludwig von Bertalanffy** in the 1940s.
- GST focuses on the interconnections and interdependencies between components of a system.
- It is widely applied in fields such as biology, engineering, economics, and social sciences.

biology
mathematics
physics
chemistry
philosophy



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- **GST** focuses on the **interconnections** and **interdependencies** between components of a **system**.
- It is widely applied in fields such as **biology**, **engineering**, **economics**, and **social sciences**.



Timeline of General Systems Theory I

- **1928:** Ludwig von Bertalanffy first proposes **organismic theory** as foundation for **GST**.
- **1940:** Bertalanffy publishes "*Der Organismus als physikalisches System*" introducing **open systems** concept.
- **1945:** Bertalanffy formally introduces **General Systems Theory** at University of Chicago.
- **1947:** **Norbert Wiener** coins the term "**Cybernetics**" at **Macy Conferences**.
- **1948:** Wiener publishes "*Cybernetics: Or Control and Communication in the Animal and the Machine*."
- **1948:** Claude Shannon publishes "*A Mathematical Theory of Communication*" founding **information theory**.

Feedback

neural networks



Timeline of General Systems Theory II

Perception

- **1954:** Society for General Systems Research founded by Bertalanffy, Kenneth Boulding, and others.
- **1956:** Jay Forrester develops *Systems Dynamics* at MIT, focusing on feedback loops and time delays.
- **1968:** Bertalanffy publishes *General System Theory: Foundations, Development, Applications*. → mathematical
- **1972:** Donella Meadows et al. publish *The Limits to Growth* using systems dynamics modeling.
- **1975:** Fritjof Capra publishes *The Tao of Physics*, bridging systems thinking and quantum physics.
- **1976:** Richard Dawkins publishes *The Selfish Gene*, applying systems concepts to evolutionary biology.



Timeline of General Systems Theory III

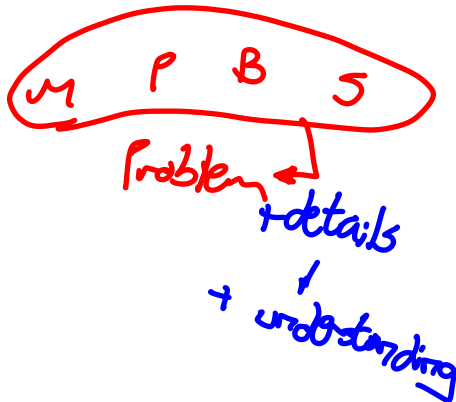
- **1977:** Ilya Prigogine wins Nobel Prize for work on dissipative structures and self-organization.
- **1980s:** Emergence of complexity science and chaos theory, heavily influenced by GST principles.
- **1984:** Santa Fe Institute founded to study complex adaptive systems.
- **1987:** Donella Meadows et al. publish Beyond the Limits, updating systems models.
- **1990:** Peter Senge publishes The Fifth Discipline, popularizing systems thinking in management.
- **1990s-2000s:** GST influences network theory, agent-based modeling, and social network analysis.
- **2004:** Donella Meadows publishes Thinking in Systems: A Primer (posthumously).



A.N.N.



General Systems Theory I



- In **general systems theory**, the **idea** is to see a **problem** from different autonomous **study areas**; it helps to create a **better full picture** of a problem or situation.
- **Systems** are **dynamic**; for that reason, you need to define **boundaries** and **constraints** to **control analysis**. Also, some systems are **highly susceptible** to changes from the environment.

↓
chaos



General Systems Theory II

- **Ludwig von Bertalanffy**

started to write a book, but he only had some ideas and insufficient background to develop them.

- He waited (twenty years) for mathematical and computational concepts to evolve, and then he was able to finish the book citing more applied concepts.

Alan Turing



General Systems Theory III

- Dynamical Systems*
complex
- In nature in the real world, everything is a system. However, the deeper you dive to understand the problem, the more complexity arises.
 - In this context, **systems theory** is useful. Some **patterns** can be detected, and some **details** can be discarded.
abstraction
- classic A.I.*
machine learning

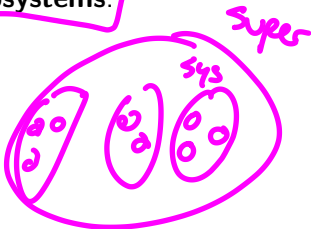


General Systems Theory IV

- **Systems hierarchies** are useful to split big problems into components, work on specific components, and then connect them as the context requires.
- A **system** can be represented by multiple internal systems. The big system is called a **super system**; internal ones are called **subsystems**.

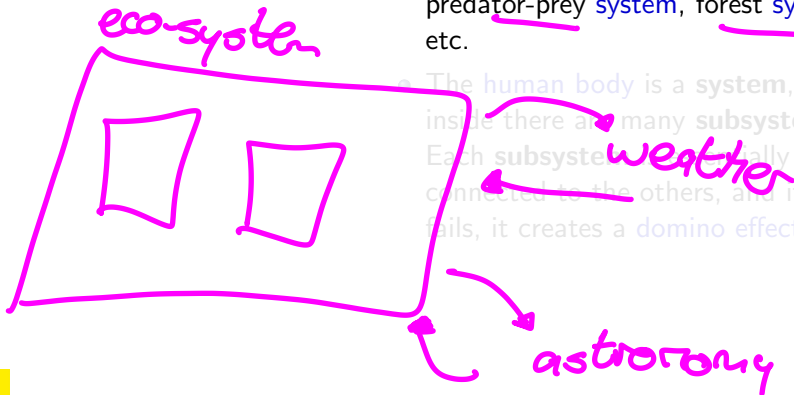
top-down → C.O.P.
architectures

responsibilities segregation



General Systems Theory V

- In nature you could think an ecosystem is a **super system** composed of different **subsystems**: water system, solar system, predator-prey system, forest system, etc.



- The human body is a system, and inside there are many subsystems. Each subsystem is usually connected to the others, and if one fails, it creates a domino effect.



General Systems Theory V

The Big Short
 economic crisis
 2007~2009

Dr. House

- In nature, you could think an **ecosystem** is a **super system** composed of different **subsystems**: water **system**, solar **system**, predator-prey **system**, forest **system**, etc.
- The **human body** is a **system**, and inside there are many **subsystems**. Each **subsystem** is essentially connected to the others, and if one fails, it creates a **domino effect**.

medicine



General Systems Theory VI

- **Everything** in the real world, in the universe **is a system**. The hardest thing is to get the right representation.

$$\nabla f$$

$$f'$$

- Remember **concepts** such as: snowball effect, butterfly effect, and domino effect. The message is the same: be careful with **details**, failures, and changes; there is **no small impact**.



General Systems Theory VI

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Chaos theory



General Systems Theory VII

- Sometimes you have an ^{Not} expected output. Most of the time, it's hard to achieve, so you must be prepared for everything.

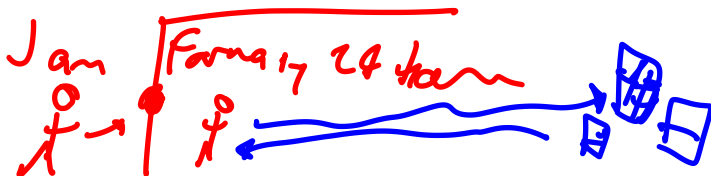
- A black box is a type of model used when you want to get the desired output based on specific input, but you don't want to expose the process used to achieve it.
↓
emergent behaviors



General Systems Theory VII



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General Systems Theory VIII

- **White-box** models are those where the **processes** are open for checking, validating, and following step by step. They are useful when you want to understand **how** the **system works**.

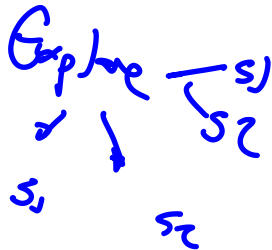


- **Critical Systems Practice** is a methodology for addressing **Critical Systems Thinking**. CSP has 4 main stages: *Explore, Produce, Intervene, and Check* — *EPIC*.



General Systems Theory VIII

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- 3 Graphs and Networks Theory
- 4 Paradigms Supporting GST



What is Information Theory?

- **Information theory** studies the **quantification**, **storage**, and **transmission** of information.
- It was founded by **Claude Shannon** in 1948.
- Key concepts include **entropy**, **mutual information**, and **channel capacity**.

metrics → mathematics



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uncertainty
↓
randomness
how much information
is pending



Entropy in Information Theory

- **Entropy** measures the uncertainty or randomness in a system.
- **High entropy**: more uncertainty → less predictability.
- **Low entropy**: less uncertainty → more predictability.
- Formula: $H(X) = - \sum p(x) \log_2 p(x)$, where $p(x)$ is the probability of event x .



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Applications of Entropy

- **Data Compression:** Reducing file sizes by removing ~~redundancy~~.
- **Cryptography:** Ensuring secure communication by maximizing entropy.
- **Systems Analysis:** Measuring the complexity and uncertainty of systems.
- **Thermodynamics:** Understanding energy distribution in physical systems.

verano → ↓

los

~ ~ ~
~ ~ ~
~ ~ ~



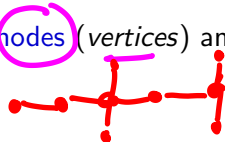
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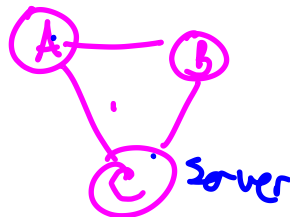
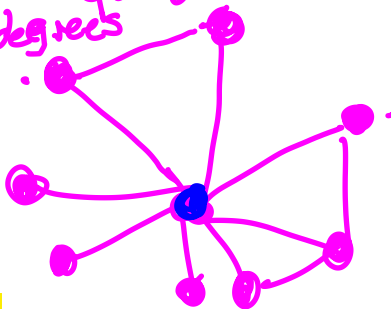


What is Graph Theory?

- **Graph theory** studies the relationships between nodes (*vertices*) and edges (*connections*).
- It was introduced by **Leonhard Euler** in 1736.
- Applications: social networks, transportation systems, computer networks.

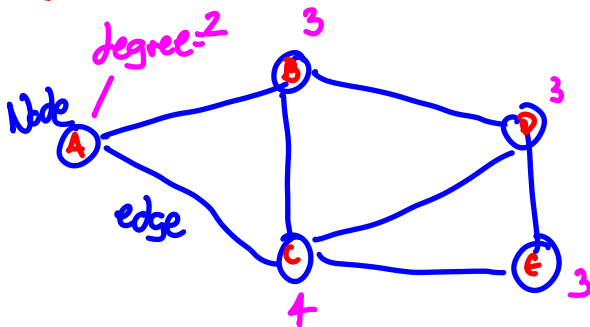


Six separation
degrees



Key Concepts in Graph Theory

- **Node (Vertex):** A **point** in the graph.
- **Edge:** A **connection** between two nodes.
- **Degree:** The **number of edges** connected to a node.
- **Path:** A **sequence of edges** connecting nodes.
- **Cycle:** A path that **starts and ends at the same node**.



Path = A-B-D-E

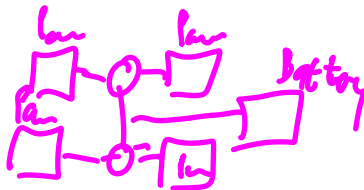
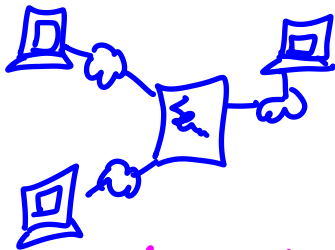
or
A-C-E

cycle =
A-B-D-E-C



What is Network Theory?

- **Network theory** extends graph theory to study **real-world systems**.
- Focuses on **structure**, **dynamics**, and **functionality**.
- Examples: **Internet**, **power grids**, **biological networks**.



Case Study: Metabolic Network

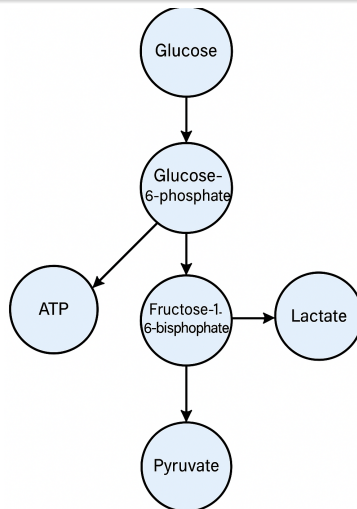


Figure: Metabolic network of *glycolysis* pathway.



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Key Paradigms in General Systems Theory

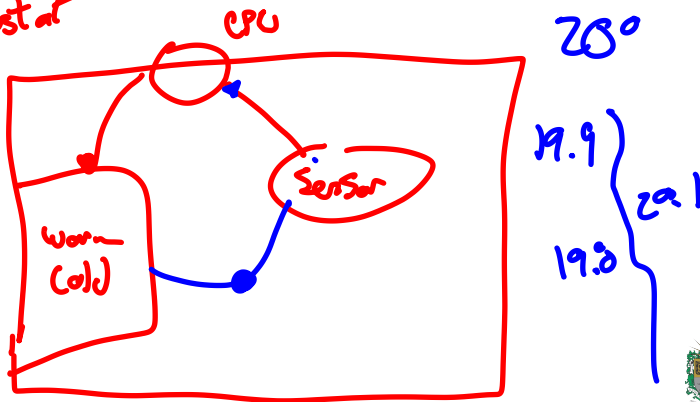
- ✓ • **Cybernetics:** Studies control and communication in systems.
- ✓ • **Systems Thinking:** Focuses on interconnections and feedback loops.
- ✓ • **Complexity Science:** Explores emergent behavior in complex systems.
- ✓ • **Systems Dynamics:** Models time-dependent behavior of systems.
- ✓ • **Agent Theory:** Studies individual agents and their interactions in systems.
 A.I.
- ✓ • **Network Theory:** Analyzes relationships and connections in systems.
- ✓ • **Cellular Automata:** Models discrete systems with simple rules.
- ✓ • **Fuzzy Logic:** Deals with uncertainty and imprecision in systems.
- ✓ • **Chaos Theory:** Studies sensitive dependence on initial conditions.
- ✓ • **Game Theory:** Analyzes strategic interactions between agents.



Cybernetics and GST

- It was founded by **Norbert Wiener** in 1948.
- Focuses on feedback, control, and adaptation.
- Applications: robotics, artificial intelligence, management systems.

Thermostat



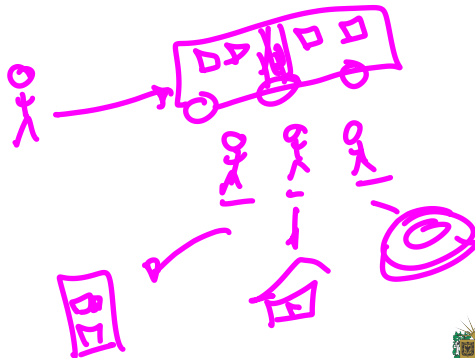
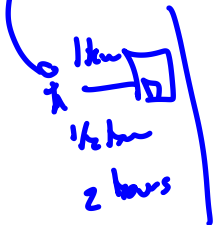
Systems Thinking and GST

- It emphasizes the holistic understanding of systems.
- Key principles: interdependence, feedback, emergence.
- Applications: organizational management, ecology, policy-making.



Complexity Science and GST

- Studies non-linear, adaptive, and emergent systems.
- Examples: ecosystems, financial markets, social systems.
- Tools: agent-based modeling, network analysis.



Systems Dynamics and GST

- It was developed by Jay Forrester in the 1950s.
- Models feedback loops and time delays.
- Applications: supply chain management, urban planning, climate change.

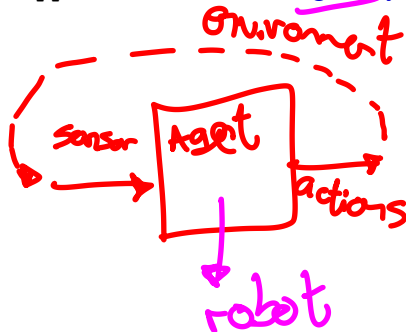
differential equations



Agent Theory and GST

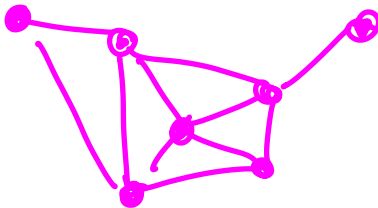
sensors
actuators

- It studies individual agents and their interactions.
- Key concepts: autonomy, adaptation, learning. → A.I.
- Applications: multi-agent systems, social networks, game theory.



Network Theory and GST

- It studies relationships and connections in systems.
- Key concepts: nodes, edges, centrality.
- Applications: social networks, transportation systems, biological networks.



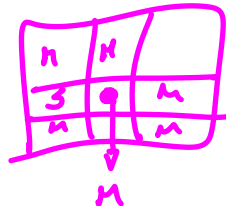
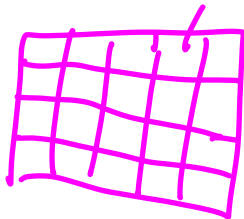
Cellular Automata and GST

- It models discrete systems with simple rules.
- Key concepts: cells, states, neighborhoods.
- Applications: pattern formation, biological systems, computer graphics.

spatial

state

Atari



$t_1 \rightarrow t_2 \rightarrow t_3 \rightarrow \dots$

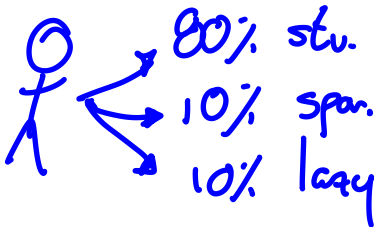
discrete

simulations



Fuzzy Logic and GST

- It deals with uncertainty and imprecision.
- Key concepts: fuzzy sets, membership functions, fuzzy rules.
- Applications: control systems, decision-making, pattern recognition.

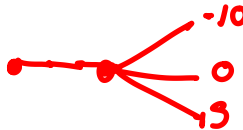


Chaos Theory and GST

- It studies sensitive dependence on initial conditions.
- Key concepts: chaotic systems, bifurcations, strange attractors.
- Applications: weather prediction, financial markets, biological systems.

productive
systems

non-linear



Game Theory and GST

- Studies strategic interactions between agents.
- Key concepts: players, strategies, payoffs.
- Applications: economics, political science, biology.

Handwritten payoff matrix for a game between two players, t_1 and t_2 .

Player t_1 strategies: Coffee, Silence

Player t_2 strategies: C, S

	Coffee	Silence
C	5	10
S	0	1

The matrix shows the payoffs for each combination of strategies. The first number in each pair is the payoff for player t_2 , and the second number is the payoff for player t_1 .



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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/systems-analysis>

